Link-State Forecasting: Final Report in One Page

Collins–ISU WiCI, Nov 2026

1. The goal

- Forecast RTT, jitter, and loss of **Starlink, OneWeb, and 5G** links **0.2–3 s ahead** (5 Hz data) for the prediction-aware multipath (MPQUIC) scheduler.
- The scheduler needs two things: a good forecast *and* a trustworthy bound on how wrong it may be.

2. What was wrong before (audit of the initial pipeline)

- **Data leak (Starlink):** missing pings were filled by interpolating toward the *next* ping, so the input already held part of the answer (11.9% of Starlink DL samples). Fixing it raises the 1-step error by 9.7%.
- **Fake calibration:** 95% intervals were computed from the test errors themselves, so coverage was 0.95 by construction.
- **Smaller issues:** horizon labeled "15 s" is really 3 s, constant targets counted as $R^2 = 1$, single training run per model.

3. What I did differently

1. **Causal data:** replaced interpolated inputs with the last real ping, for every method (initial model retrained on the same data, so the comparison is fair).
2. **Anchored decoder:** the model starts from the last measured value and predicts the change.
3. **Distribution instead of one number (the key change):** each output gives a mean and 5 quantiles. Reason: RTT is spiky (RMSE is 1.5–3.8$\times$ the MAE), so the mean is pulled up by spikes. Median $\rightarrow$ best for absolute error, quantiles $\rightarrow$ uncertainty.
4. **Honest calibration:** intervals corrected on held-out sessions (conformal), coverage measured on unseen test sessions. 3-seed ensembles.
5. **Fair evaluation:** 3 seeds, 4 simple baselines (persistence, 1 s repeat, two ridge models), per-session significance test, 80 extra sensitivity runs.

4. Results (final vs. initial model, test sessions, RTT, 3 seeds)

	OneWeb DL	OneWeb UL	Starlink DL	Starlink UL	5G DL	5G UL
95% interval score	-19.0%	-24.0%	-39.3%	-8.4%	-13.3%	-28.1%
MAE	-9.8%	-2.4%	-10.9%	-2.8%	-1.5%	-17.7%
RMSE	-2.2%	0.4%	-4.1%	0.1%	-1.0%	-0.3%

negative = better. Interval coverage on unseen sessions: 0.935–0.954 (target 0.95).

- **Uncertainty:** 95% interval score **-22.0% on average** (lower is better), all 6 links (e.g. Starlink DL: 100 ms wide instead of 176 ms, same coverage).
- **Accuracy:** RTT error (MAE) **-7.5% on average**, all 6 links. RMSE unchanged.
- **Significant:** final model better in 142 of 156 test sessions, $p < 0.01$ on 5 of 6 links.
- **Where it comes from:** only from the distributional output (step 3). Steps 1–2 alone did not lower the MAE.
- **Robust:** holds for 3 s and 12 s input windows and a 6 s horizon. A quantile-loss weight of 4 improves it further (OneWeb DL MAE -10%).
- **Cost:** 10.0 ms per forecast on one CPU thread (budget 200 ms), 4% more parameters.

5. Takeaways and limits

- The gain is **calibrated, input-dependent uncertainty**: tight bounds when a path is calm, wide before bursts $\rightarrow$ the scheduler duplicates packets only when needed.
- **Limits:** RMSE did not improve. A tuned linear model is close on RMSE. One site, one data split. Not yet tested end-to-end in the tunnel (next step).
- **Delivered:** final report (53 pp.), IEEE TWC paper draft, 30-slide deck, code, models, and data on GitHub.